



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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Lecture Notes

22GE004 – BASICS OF ELETRONICS ENGINEERING

UNIT 4 – PN JUNCTION AND BJT AS SWITCH

Introduction

Electronic circuits often require the ability to control the flow of current. Two fundamental components play a crucial role in achieving this: the PN junction diode and the Bipolar Junction Transistor (BJT). While they operate on different principles, both can function as electronic switches.

PN Junction Diode as a Switch

- **PN Junction:** A diode is formed by joining two doped regions in a semiconductor material - a p-type region and an n-type region. This creates a barrier called the depletion region.
- **Operation as a Switch:**
 - **Forward Bias:** When a positive voltage is applied to the p-type region (anode) and a negative voltage to the n-type region (cathode), the depletion region narrows. This allows significant current flow in the forward direction. The diode acts as a closed switch.
 - **Reverse Bias:** When the biasing is reversed (positive to cathode and negative to anode), the depletion region widens. This significantly reduces current flow. The diode acts as an open switch.
- **Limitations:** Diodes are unidirectional switches, allowing current flow only in the forward direction.

Bipolar Junction Transistor (BJT) as a Switch

- **Construction:** BJT is a three-terminal device consisting of two PN junctions and three regions: base, collector, and emitter.
- **Operation as a Switch:**
 - **Saturation Mode:** By applying a small current to the base terminal, a much larger current can flow between the collector and emitter. The BJT acts as a closed switch with a high current gain.
 - **Cut-off Mode:** When no current is applied to the base, the collector-emitter current is very small. The BJT acts as an open switch.
- **Advantages:** BJTs offer higher current handling capacity compared to diodes and can be controlled with a small base current, making them more efficient switches.

Key Differences

Feature	PN Junction Diode	BJT
Construction	Two PN junctions	Three regions, two PN junctions
Biasing	Unidirectional	Bidirectional
Control	Voltage	Current
Current Gain	Low	High

Summary

Both PN junction diodes and BJTs can be used as electronic switches. However, they have distinct characteristics. Diodes are simpler, unidirectional switches, while BJTs offer higher current gain, bidirectional control, and greater efficiency. The choice between these components depends on the specific application requirements.